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|---|---|
| **Started on** | Tuesday, 29 September 2026, 7:05 PM |
| **State** | Finished |
| **Completed on** | Tuesday, 29 September 2026, 7:38 PM |
| **Time taken** | 33 mins 29 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:
The first line contains the no of elements in the list-n
The next n lines contain the elements.

Output:
Sorted list of elements

**For example:**

| Input | Result |
|---|---|
| 5<br>67 34 12 98 78 | 12 34 67 78 98 |

**Answer:**

```c
#include<stdio.h>
void quicksort(int a[],int low,int high){
    if(low<high){
        int pivot=a[high];
        int i=low-1;
        for(int j=low;j<high;j++){
            if(a[j]<pivot){
                i++;
                int temp=a[i];
                a[i]=a[j];
                a[j]=temp;
            }
        }
        int temp=a[i+1];
        a[i+1]=a[high];
        a[high]=temp;
        int p=i+1;
        quicksort(a,low,p-1);
        quicksort(a,p+1,high);
    }
}
int main(){
    int n;
    scanf("%d",&n);
    int a[n];
    for(int i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    quicksort(a,0,n-1);
    for(int i=0;i<n;i++){
        printf("%d ",a[i]);
    }
    return 0;
}

```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 5<br>67 34 12 98 78 | 12 34 67 78 98 | 12 34 67 78 98 | ✔ |
| ✔ | 10<br>1 56 78 90 32 56 11 10 90 114 | 1 10 11 32 56 56 78 90 90 114 | 1 10 11 32 56 56 78 90 90 114 | ✔ |
| ✔ | 12<br>9 8 7 6 5 4 3 2 1 10 11 90 | 1 2 3 4 5 6 7 8 9 10 11 90 | 1 2 3 4 5 6 7 8 9 10 11 90 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.